

AUGSBURG, Germany

WMO: 108520

Lat: 48.4253N

Lon: 10.9419E

Elev: 1515

StdP: 13.91

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	13.9	2.8	6.7	10.2	8.3	8.9	15.2	29.1	42.8	25.4	41.1	4.8	60	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.9	86.0	67.7	82.3	65.9	79.0	64.6	69.1	81.5	67.4	78.7	66.0	76.3	6.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	97.3	72.9	63.5	92.8	71.4	62.3	88.9	70.1	34.0	81.7	32.7	79.2	31.6	76.4	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.3	18.9	16.1	2.9	90.9	9.4	2.9	-3.9	92.9	-9.3	94.6	-14.6	96.2	-21.4	98.3		
(5)			2.4	72.0	9.3	1.5	-4.3	73.1	-9.7	74.0	-15.0	74.8	-21.8	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	32.2	31.4	39.9	47.2	55.0	61.2	64.5	65.2	56.9	49.0	38.8	34.3
	DBStd	13.89	8.23	9.76	7.65	7.08	6.65	6.57	5.13	5.96	5.66	6.43	6.10	7.79
	HDD50	2487	551	521	319	134	33	3	0	0	6	96	338	486
	HDD65	6361	1016	941	779	535	314	149	73	73	249	496	785	951
	CDD50	1782	0	0	6	49	187	338	449	471	214	65	2	1
	CDD65	182	0	0	0	0	3	35	57	79	8	0	0	0
	CDH74	2050	0	0	0	21	49	385	638	859	97	1	0	0
	CDH80	514	0	0	0	2	1	88	153	259	11	0	0	0
Wind	WSAvg	6.4	7.8	6.8	7.9	6.7	6.2	5.5	5.5	5.1	5.1	5.8	6.0	7.7
Precipitation	PrecAvg	31.9	1.9	1.7	1.7	2.4	3.4	4.1	3.6	3.7	2.5	2.2	2.4	2.0
	PrecMax	41.5	5.2	5.5	4.5	4.5	7.6	9.3	7.2	7.3	5.1	6.5	6.4	4.5
	PrecMin	25.6	0.5	0.2	0.4	0.8	0.6	1.9	1.2	1.0	0.4	0.1	0.6	0.4
	PrecStd	4.3	1.0	1.0	1.0	1.1	1.7	1.6	1.6	1.5	1.3	1.6	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	60.2	67.5	76.9	79.2	89.7	89.9	91.2	82.2	72.8	58.8	54.8
		MCWB	47.2	47.4	52.1	58.6	62.7	69.4	68.4	68.8	65.5	59.5	51.4	49.6
	2%	DB	48.5	52.3	61.0	70.2	75.0	82.6	84.4	86.3	77.2	68.1	53.6	51.4
		MCWB	44.0	45.4	49.0	55.0	59.8	66.4	66.4	67.6	62.9	58.1	48.9	46.9
	5%	DB	45.6	47.8	55.9	66.0	71.9	78.8	80.9	82.6	72.9	63.5	50.3	48.2
		MCWB	42.1	43.3	47.1	53.3	57.8	64.8	65.2	66.3	61.9	55.8	46.9	44.8
	10%	DB	42.8	43.1	51.5	60.7	69.0	75.1	77.4	78.9	68.6	59.1	48.1	44.8
		MCWB	40.1	39.8	45.2	50.8	56.5	63.3	64.0	65.1	59.9	54.3	45.5	42.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	48.6	53.2	59.9	65.1	71.7	70.2	71.7	67.0	61.1	52.6	50.3
		MCDB	52.0	56.2	65.6	74.5	76.5	85.2	82.5	84.6	77.5	70.0	55.6	53.6
	2%	WB	44.9	46.1	50.0	56.5	62.1	68.2	68.2	69.3	64.8	58.5	49.6	47.5
		MCDB	47.8	51.8	58.0	67.9	71.2	79.3	80.4	81.9	74.2	65.5	52.8	50.7
	5%	WB	42.3	43.3	47.9	54.1	59.5	66.4	66.8	67.5	63.0	56.6	47.7	45.0
		MCDB	44.6	47.3	54.4	63.8	68.5	76.4	77.8	79.1	70.6	62.5	50.1	47.9
	10%	WB	40.4	40.1	46.1	51.6	57.4	64.2	65.3	65.8	60.9	54.5	45.5	42.2
		MCDB	42.5	42.7	51.4	59.4	66.4	73.4	74.9	76.4	66.9	58.8	47.6	44.3
Mean Daily Temperature Range	5% DB	MDBR	9.6	13.1	15.3	17.7	19.4	18.5	19.8	20.9	18.5	15.5	10.5	9.3
		MCDBR	13.1	20.7	25.3	28.2	25.3	26.3	27.3	28.4	25.9	23.7	15.6	13.8
		MCWBR	9.7	14.4	14.9	15.0	11.9	12.6	11.9	11.7	13.4	14.8	11.1	10.5
	5% WB	MCDBR	12.0	19.6	21.6	25.9	22.6	24.1	23.7	25.2	21.4	22.1	14.6	12.5
		MCWBR	9.5	14.1	13.7	14.2	11.5	12.2	11.0	11.6	11.7	14.8	10.9	10.0
Clear-Sky Solar Irradiance	taub	0.300	0.323	0.357	0.389	0.398	0.403	0.406	0.401	0.377	0.350	0.320	0.295	
	taud	2.453	2.386	2.335	2.271	2.277	2.296	2.280	2.316	2.362	2.430	2.469	2.460	
	Ebn at Noon	246	263	271	272	274	273	270	266	261	250	235	232	
	Edh at Noon	20	27	33	39	41	40	40	37	32	26	20	18	
All-Sky Solar Radiation	RadAvg	317	590	954	1368	1593	1785	1706	1492	1071	662	349	254	
	RadStd	43	81	120	181	182	150	151	145	129	79	43	31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	+0.61	+0.70	N/A	N/A	+106	+36

Nomenclature: See separate page